# 网络架构
## ONTAP 9

NetApp
January 14, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/concepts/network-connectivity-concept.html on January 14, 2022. Always check docs.netapp.com for the latest.

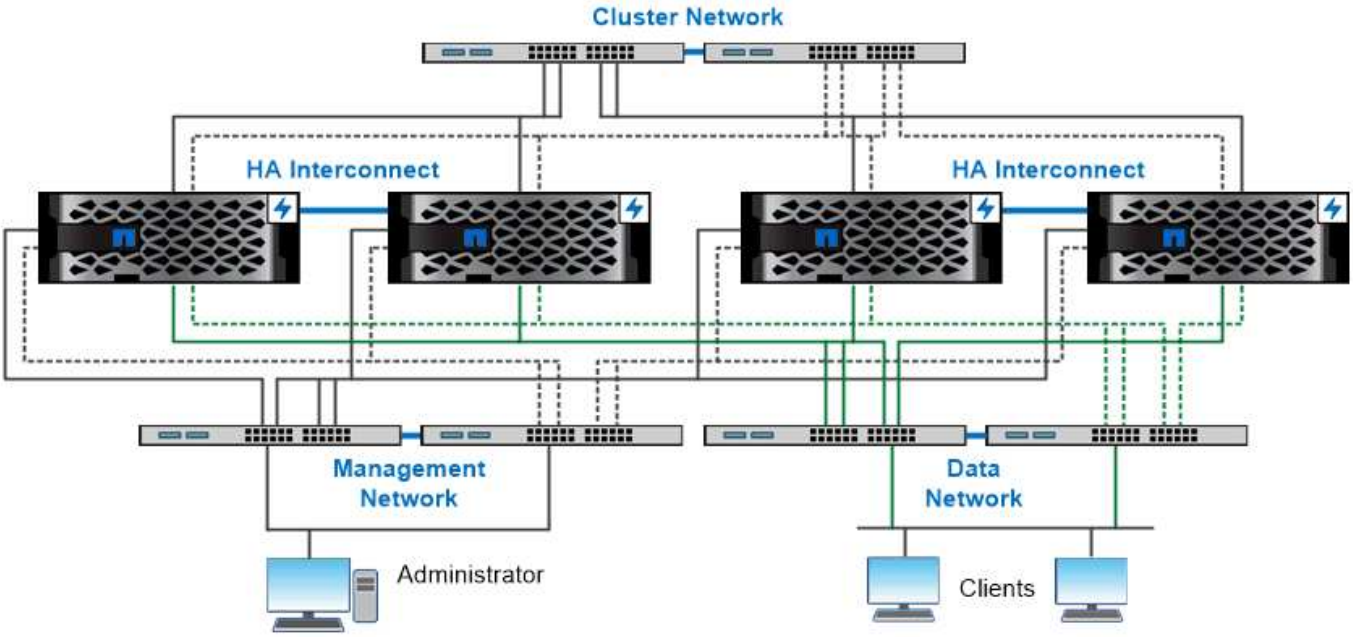
# 目录

网络架构 1
网络架构概述 1
逻辑端口 1
支持行业标准网络技术 2

# 网络架构

## 网络架构概述

ONTAP 数据中心实施的网络架构通常由集群互连，用于集群管理的管理网络和数据网络组成。NIC （网络接口卡）为以太网连接提供物理端口。HBA （主机总线适配器）为 FC 连接提供物理端口。




*The network architecture for an ONTAP datacenter implementation typically consists of a cluster interconnect, a management network for cluster administration, and a data network.*

## 逻辑端口

除了每个节点上提供的物理端口之外，您还可以使用 _logical ports_来 管理网络流量。逻辑端口是接口组或 VLAN 。

### 接口组

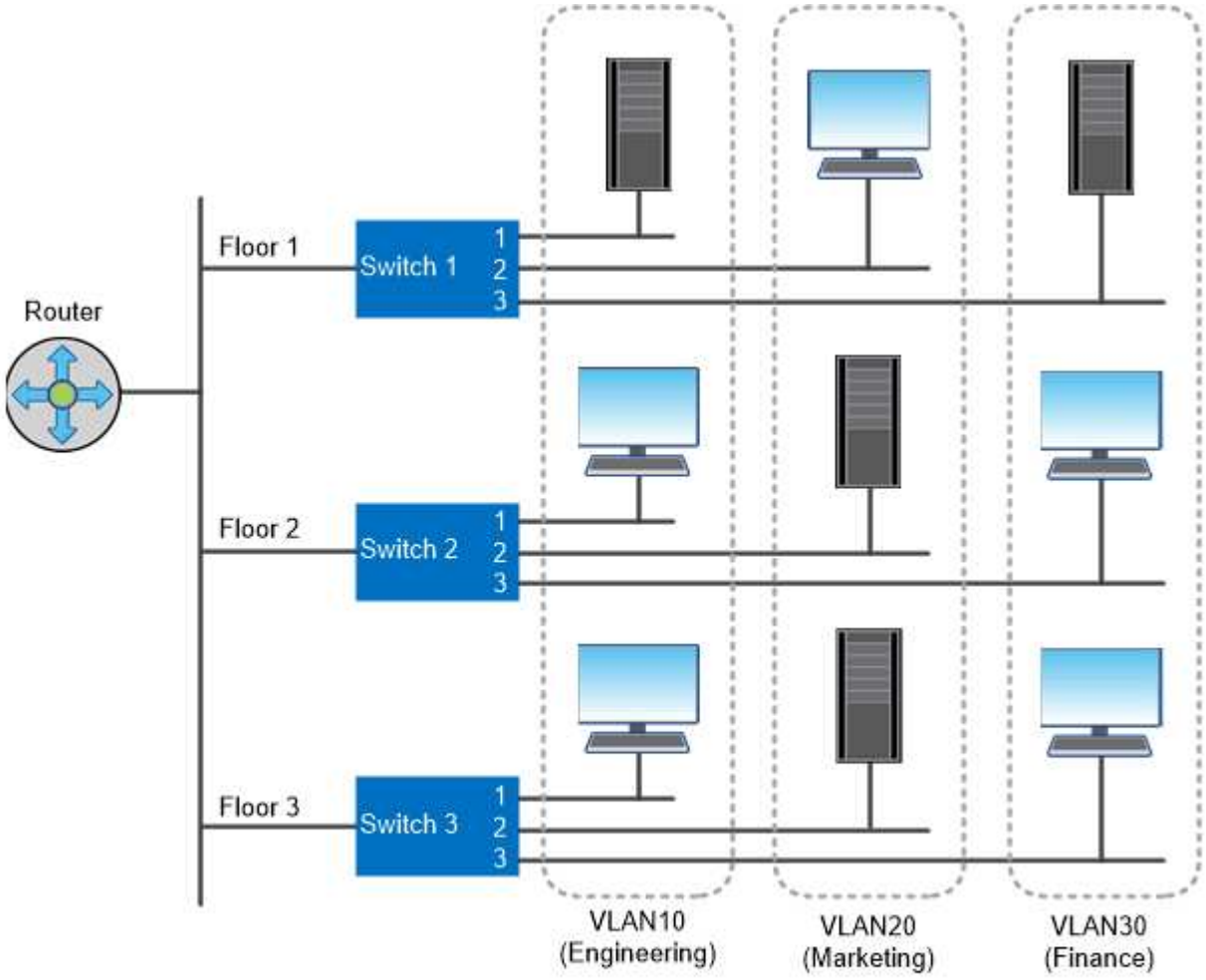
*Interface groups* 将多个物理端口组合成一个逻辑 "`trunk port" 。` 您可能需要创建一个由不同 PCI 插槽中 NIC 的端口组成的接口组，以确保不会因插槽故障而导致业务关键型流量中断。

接口组可以是单模式，多模式或动态多模式。每个模式提供不同级别的容错。您可以使用任一类型的多模式接口组来对网络流量进行负载平衡。

### VLAN

*VLANs* 将流量从网络端口（可能是接口组）分离到基于交换机端口定义的逻辑分段中，而不是物理边界。属于 VLAN 的 _end-station_are related by function or application 。

您可以按部门（如工程和营销）或项目（如释放 1 和释放 2 ）对终端工作站进行分组。由于终端工作站的物理位置与 VLAN 无关，因此终端工作站可能位于地理位置较远程的位置。




*You can use VLANs to segregate traffic by department.*

# 支持行业标准网络技术

ONTAP 支持所有主要的行业标准网络技术。关键技术包括 IP 空间， DNS 负载平衡和 SNMP 陷阱。

中介绍了广播域，故障转移组和子网 NAS 路径故障转移。

## IP 空间

您可以使用 *ipspaces* 为集群中的每个虚拟数据服务器创建一个不同的 IP 地址空间。这样，在管理上独立的网络域中的客户端就可以访问集群数据，同时使用来自同一 IP 地址子网范围的重叠 IP 地址。

例如，服务提供商可以为使用相同 IP 地址的租户配置不同的 IP 空间来访问集群。

## DNS 负载平衡

您可以使用 _DNS 负载平衡 _ 在可用端口之间分布用户网络流量。DNS 服务器会根据接口上挂载的客户端数量动态选择用于流量的网络接口。

## SNMP 陷阱

您可以使用 _SNMP 陷阱 _ 定期检查操作阈值或故障。SNMP 陷阱捕获从 SNMP 代理异步发送到 SNMP 管理器的系统监控信息。

**_FIPS 兼容性 _**

对于所有 SSL 连接， ONTAP 均符合联邦信息处理标准（ FIPS ） 140-2 的要求。您可以打开和关闭 SSL FIPS 模式，全局设置 SSL 协议以及关闭 RC4 等任何弱密码。

**Copyright Information**

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

**Trademark Information**

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.